

Matching for Causal Inference Without Balance Checking

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joint work with
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(talk at the Society for Political Methodology, University of Michigan, 7/12/08)

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- (Medical experiments are the reverse: small- n with random treatment assignment; don't match unless something goes wrong)

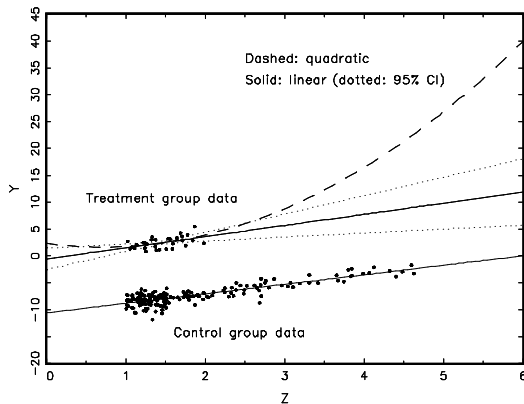
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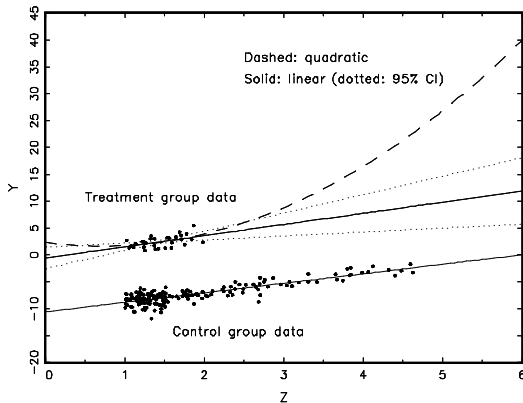
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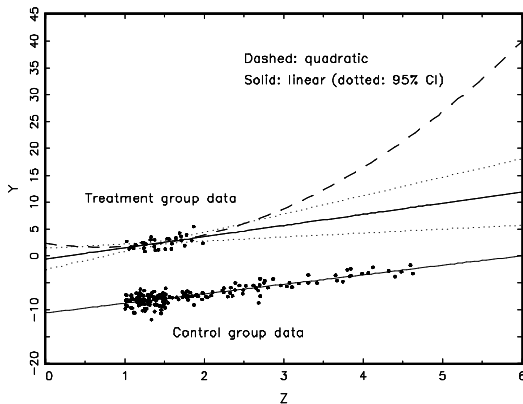
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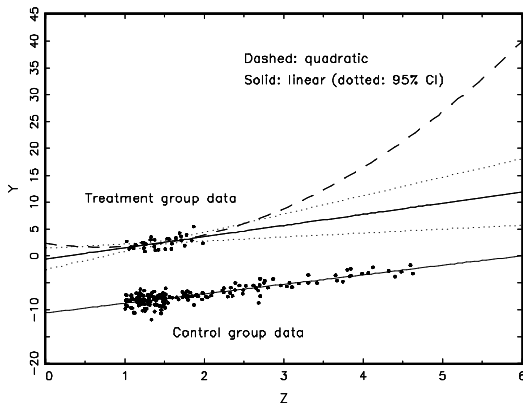


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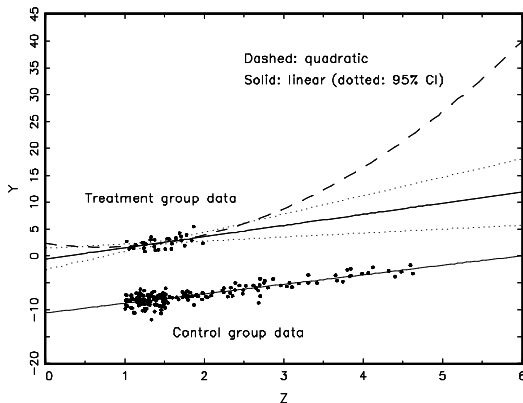


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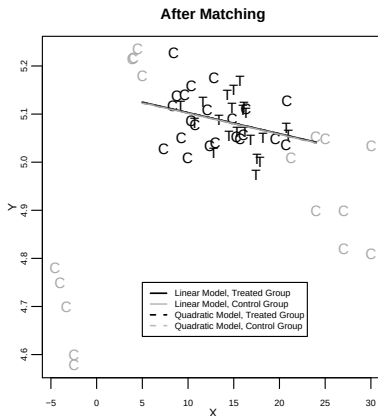
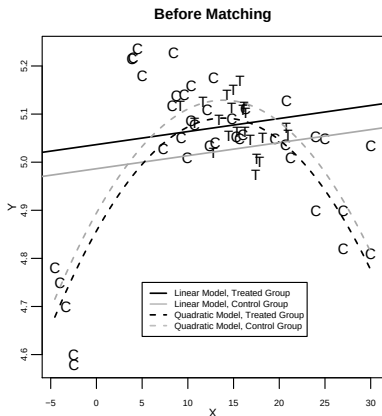
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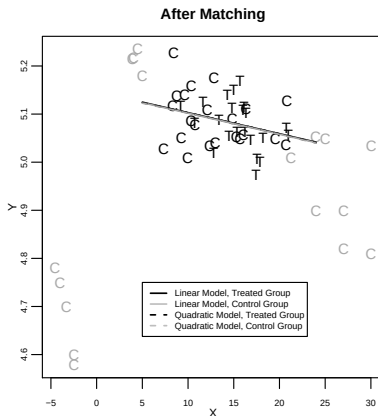
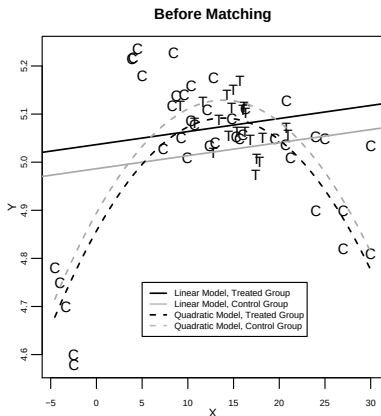
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Matching reduces model dependence, bias, and variance

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- All methods attempt to achieve good **balance** between treated and control groups, so that X is unimportant

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(Is balance even improved?)

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“Imbalance” given chosen distance metric

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Easy to adjust vector of tuning parameters, one for each X

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If ϵ_J is reduced, $\gamma_J(\epsilon_J)$ won't increase & $\gamma_H(\epsilon_H)$ is unchanged

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⇒ CEM dominates EPBR-methods in EPBR Data

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Monte Carlo: Exact replication of Diamond and Sekhon (2005), using data from Dehejia and Wahba (1999). CEM coarsening automated.

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⇒ CEM works well in non-EPBR data too

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 - & whatever else you all come up with.

For more information

<http://GKing.Harvard.edu/cem>